

Smart Pharmacy Supply Chain System

Digital Medicine Batch Tracking, FEFO Dispensing & Cold-Chain Monitoring

Student Details

Student 1: Shudiksha Sugeerthanaa
Register Number: 44731123

Student 2: Sofiya S
Register Number: 44731112

Team ID: CSE-AI-CAP-2026

Academic Details

Department:
B.E. Computer Science & Engineering (AI)

Institution:
Sathyabama Institute of Science and Technology (SIST)

Academic Year:
2026 - 2027

What is a Pharmacy Supply Chain?

A pharmacy supply chain is the step-by-step process of how medicines travel safely from pharmaceutical manufacturers to hospital pharmacies and patients.

1. Manufacturer

Produces pharmaceutical drugs in discrete batches with fixed manufacturing and expiry dates.

2. Distributor / Supplier

Transports medicine boxes to hospitals while maintaining safe temperature standards.

3. Hospital Pharmacy

Stores stock safely, checks expiry, and dispenses the right medicine batch to patients.

Why is Proper Tracking Important?

Patient Safety: Patients must never receive an expired, compromised, or counterfeit drug.

Drug Potency: Temperature-sensitive medicines like Insulin lose strength if stored in the wrong temperature.

Preventing Losses: Using medicines in the correct order prevents medicines from expiring on pharmacy shelves.

Real-World Pharmacy Inefficiencies

Hospital pharmacies face 5 major practical challenges that cause medical and financial losses:

1. High Medicine Expiry Wastage

Medicines often sit unnoticed in storage racks and expire before staff realizes they need to be used.

2. Random Dispensing Order

Staff naturally pick the newest boxes in front, leaving older batches behind to expire unnoticed.

3. Unmonitored Cold-Chain Storage

Medicines like Insulin or Vaccines spoil if refrigerators get too warm, but temperature checks are rarely automated.

4. Counterfeit Medicine Risks

Unverified or fake medicine batches can enter the hospital supply chain without a quick verification tool.

5. Manual Inventory Errors

Paper-based stock logs lead to human counting mistakes, stock-outs during emergencies, and missing batches.

The Traditional Manual Approach

Most small and medium pharmacies still rely on paper registers or basic Excel spreadsheets.

Pharmacy Task	Existing Manual Method	Major Bottleneck / Risk
Stock Recording	Writing medicine entries by hand in paper books.	Handwriting errors, lost pages, slow searches.
Expiry Tracking	Staff manually read tiny printed dates on boxes every month.	Easy to miss boxes; expired stock given to patients.
Medicine Dispensing	Pharmacist grabs whichever bottle is nearest on shelf.	Violates FEFO; older stock expires on the shelf.
Temperature Checks	Manual thermometer readings logged on a clipboard.	No instant alerts when a refrigerator fails at night.
Batch Verification	Phone calls or paper invoice checks with suppliers.	Slow process; counterfeit drugs cannot be caught fast.

Our Centralized 3-Tier System

We built a clean, centralized web application that automates pharmacy operations in 3 simple tiers:

Tier 1: Frontend
HTML5 / CSS3 / Vanilla JS
Clean UI screen staff uses

-
>

Tier 2: Backend
Java Spring Boot 3 REST API
Executes FEFO & check logic

-
>

Tier 3: Database
MySQL 8.0 Relational DB
Safely stores all pharmacy data

What Our System Does Automatically

- Picks the earliest expiring batch automatically when dispensing (FEFO).
- Calculates expiry alerts at 90, 60, and 30 days before expiry.
- Locks expired and rejected batches so they cannot be dispensed.

Key Safety Guarantees

- Instantly flags cold-chain temperature readings that are outside safety limits.
- Checks batch numbers against authentic records to stop counterfeit drugs.
- Keeps an itemized digital receipt for every medicine dispensed.

SMART Project Objectives

Our project was developed with clear, measurable goals designed for real hospital pharmacies:

1. Digital Batch Tracking

Track every physical batch with unique batch number, medicine name, manufacturer, supplier, and expiry date.

2. Strict FEFO Dispensing

Enforce 100% First-Expired-First-Out dispensing logic so older authentic batches are always issued first.

3. Multi-Tier Expiry Radar

Categorize inventory into Expired, 30-day (Critical), 60-day (Warning), and 90-day (Notice) alert windows.

4. Cold-Chain Compliance Checking

Validate thermal sensor readings against medicine temperature ranges (e.g. 2C to 8C for Insulin).

5. Anti-Counterfeit Batch Verification

Verify whether a batch is VERIFIED, UNVERIFIED, REJECTED, or NOT FOUND with 1-click verify and reject actions.

8 Key Features Implemented

1. Medicines

Add and edit catalog drugs with brand name, dosage, and min/max temperature limits.

2. Suppliers

Maintain verified manufacturer and vendor contacts, phones, and warehouse addresses.

3. Batches

Track discrete batches with unique IDs, quantities, and manufacturing & expiry dates.

4. FEFO Dispense

Core engine: automatically selects earliest expiring batch and updates stock balances.

5. Expiry Radar

Alerts tab filtering batches into Expired, 30 days, 60 days, and 90 days remaining.

6. Temp Logger

Evaluates sensor readings: flags COMPLIANT or OUT_OF_RANGE with diagnostic alerts.

7. Verification

Scans batch numbers to detect counterfeit drugs with 1-click Verify/Reject controls.

8. Dashboard

5 live KPI summary counters and 3 real-time alert tables without heavy charts.

Technology Stack Breakdown

Frontend (Client Tier)

â€¢ **HTML5:** Semantic layout structure (<header>, <nav>, <main>, <table>).

â€¢ **CSS3:** Custom clinical design system using CSS Grid & Flexbox (no Tailwind, no Bootstrap).

â€¢ **Vanilla JavaScript (ES6+):** Clean browser logic without heavy React or Angular dependencies.

â€¢ **Fetch API:** Asynchronous JSON HTTP calls connecting to the backend.

Backend (Application Tier)

â€¢ **Java 17:** Modern, secure, object-oriented programming language.

â€¢ **Spring Boot 3:** Lightweight, production-ready framework for REST APIs.

â€¢ **Spring Data JPA:** Hibernate ORM for safe, boilerplate-free database operations.

â€¢ **Bean Validation:** Validates clean inputs (@NotNull, @NotBlank, @Positive).

Database (Storage Tier)

MySQL 8.0: Robust relational database engine managing 3NF normalized tables with foreign keys and unique constraints. (Includes local H2 fallback for instant viva demonstration).

Developer & Testing Tools

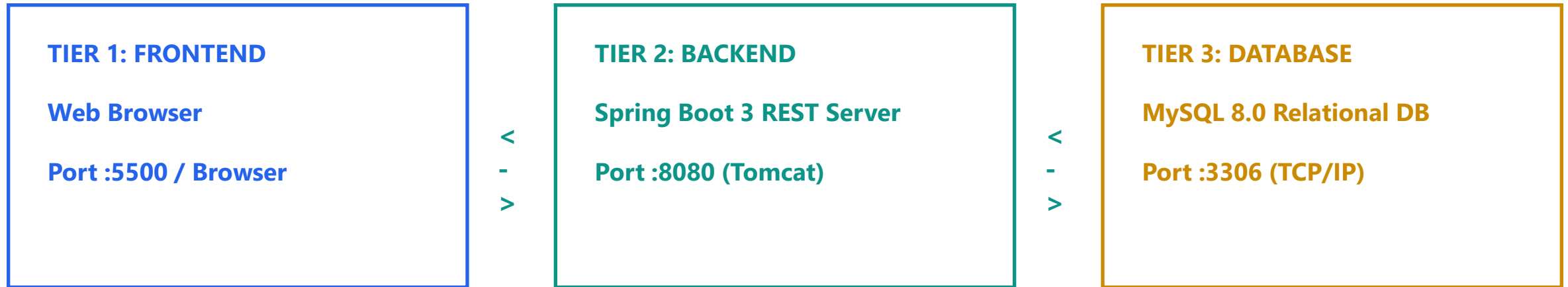
Antigravity IDE / VS Code: Code editing & debugging.

Apache Maven: Dependency & build management.

Postman: REST API endpoint testing.

3-Tier Architecture & Multi-Port Data Flow

The system follows a strict, industry-standard 3-tier architecture with dedicated network ports:



Step-by-Step Data Flow:

Pharmacy Staff clicks 'Dispense' on screen -> Frontend sends JSON to Backend (Port 8080) -> Spring Boot checks FEFO rules and executes SQL on MySQL (Port 3306) -> MySQL returns updated records -> Frontend displays the instant receipt!

Application Modules Breakdown

8 dedicated functional modules covering complete pharmaceutical inventory management:

Module Name	Web Page File	Single-Sentence Purpose
1. Dashboard	index.html	Displays 5 quick summary numbers and recent alert tables at a single glance.
2. Medicines	medicines.html	Allows staff to register, update, and search medicines and their safe storage temperature limits.
3. Suppliers	suppliers.html	Stores certified manufacturer contact details, emails, and physical warehouse addresses.
4. Batches	batches.html	Tracks discrete medicine batches with unique batch numbers, stock count, and expiry dates.
5. FEFO Dispense	dispense.html	Automatically selects and deducts from the earliest-expiring verified batch to prevent wastage.
6. Expiry Alerts	expiry.html	Groups inventory into Expired, 30-day, 60-day, and 90-day alert tabs so near-expiry stock is quarantined.
7. Temperature Logger	temperature.html	Logs thermal sensor readings and checks if cold-chain medicine temperatures stay within safe limits.
8. Batch Verification	verification.html	Looks up batch authenticity and lets pharmacists mark suspect stock as VERIFIED or REJECTED.

Normalized MySQL 3NF Database Structure

The database is normalized to Third Normal Form (3NF). This avoids repeating data and guarantees that every table has a Primary Key and clean Foreign Keys.

medicines Table

Columns: id (PK), name (UNIQUE), generic_name, category, dosage, min_temperature, max_temperature, created_at.

Purpose: Stores the official drug catalog and storage rules.

suppliers Table

Columns: id (PK), supplier_name (UNIQUE), contact_person, phone, email, address, created_at.

Purpose: Stores authorized distributor credentials.

batches Table

Columns: id (PK), batch_number (UNIQUE), medicine_id (FK), supplier_id (FK), manufacturer, manufacturing_date, expiry_date, quantity, storage_requirement, verification_status.

Purpose: Physical inventory ledger used for FEFO sorting.

temperature_logs Table

Columns: id (PK), batch_id (FK), recorded_temperature, recorded_at, compliance_status, notes.

Purpose: Stores cold-chain temperature readings.

Simple Relationships: 1 Medicine -> has many Batches | 1 Supplier -> provides many Batches | 1 Batch -> has many Temperature Logs

Data Flow Diagrams (Level 0 & Level 1)

DFD Level 0 (Context Level)

[External Entity]

Pharmacy Staff / Admin

|

v (Inputs & Requests)

[Main System Process]

[0.0] Smart Pharmacy Supply Chain System

|

v (SQL Reads / Writes)

[Data Store]

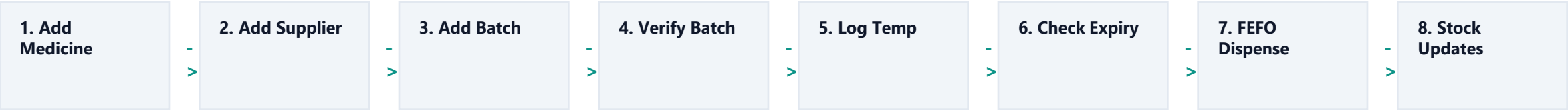
MySQL 8.0 Database

DFD Level 1 (Major Processes)

- 1.0 Manage Drugs: Add/Update drug names & temperature rules.
- 2.0 Manage Suppliers: Add/Update certified vendor information.
- 3.0 Manage Batches: Record physical batch number, mfg & exp date.
- 4.0 FEFO Dispensing: Query earliest verified batch & deduct stock.
- 5.0 Verify Batch: Authenticate batch & mark VERIFIED or REJECTED.
- 6.0 Cold-Chain Monitor: Compare sensor reading with drug thresholds.
- 7.0 Expiry Radar: Group active inventory by days remaining.

End-to-End Pharmacy Workflow & FEFO Example

The system guides the pharmacy staff through 8 clear, sequential steps:



Easy FEFO Concept Explained

FEFO Rule: 'The batch that expires first must be used first.'

[Batch A]: Batch A: Expires 10-October-2026 (Qty: 20) -> Status: Selected 1st (Earliest Expiry)

[Batch B]: Batch B: Expires 15-December-2026 (Qty: 40) -> Status: Selected 2nd (Expires Later)

When dispensing 10 units of Paracetamol, our system automatically deducts 10 from Batch A and leaves Batch B untouched.

Accessible Forms & Clean User Interface

The user interface is built with Vanilla HTML5 and CSS3 to ensure clean readability on any projector or screen:

Fixed Navigation

A clean left sidebar gives staff quick 1-click access to all 8 pharmacy sections without getting lost.

Fully Responsive

CSS Grid and Flexbox automatically adapt from 360px mobile viewports to 1920px widescreen clinic desktops.

Clear Visual Badges

Color-coded tags (Green for Verified, Amber for Unverified, Red for Out-of-Range or Rejected) guide quick clinical decisions.

Real-Time FEFO Receipt Card

Immediately after clicking Dispense, the screen displays a dashed receipt card showing the exact batch number used and how many units remain.

Non-Intrusive Toast Alerts

Success and warning messages pop up at the top right for 4 seconds, confirming actions without disruptive browser alert boxes.

Controller, Service & Repository Layers

The Java Spring Boot backend separates code into 3 distinct responsibilities:

1. Controller Layer
@RestController
Receives requests from browser

-
>

2. Service Layer
@Service + @Transactional
Performs FEFO & validation logic

-
>

3. Repository Layer
@Repository (Spring Data JPA)
Executes safe queries on MySQL

Real Execution Example:

Staff requests to dispense 10 Paracetamol -> DispenseController receives HTTP POST -> DispenseServiceImpl calls findAvailableBatchesForFefo, sorts by earliest expiry date, deducts stock, and saves DispenseRecord -> Returns clean JSON!

Global Exception Handler (@RestControllerAdvice)

Catches errors automatically: returns 404 if batch not found, 409 if stock is insufficient or duplicate batch is submitted, and 400 for invalid dates.

Database CRUD Operations via JPA

SQL Injection Protection: All database queries use Spring Data JPA parameterized inputs. No raw SQL strings are concatenated, protecting the hospital pharmacy from SQL injection attacks.

CRUD Action	Pharmacy Operation	Java / JPA Execution	Database Effect
C - Create	Register new medicine or add batch	medicineRepository.save(medicine)	INSERT INTO medicines ...
R - Read	Query batches in FEFO order	findAvailableBatchesForFefo()	SELECT ... ORDER BY expiry_date ASC
U - Update	Deduct stock; change verification status	batch.setQuantity(newQty)	UPDATE batches SET quantity = ?
D - Delete	Remove old or canceled records	medicineRepository.deleteById(id)	DELETE FROM medicines WHERE id = ?

Acceptance Criteria Verification Results

Every functional requirement requested for this project was tested and passed directly on the live backend:

Acceptance Criterion	Test Performed	Status Code	Result
Track Medicine Batches	Added batch with manufacturer, mfg date, exp date, and quantity.	201 CREATED	PASS
FEFO Dispensing	Dispensed Paracetamol: batch expiring in Oct chosen over Dec.	200 OK	PASS
Multi-Batch Split	Dispensed 25 units: 10 units from Batch A, 15 from Batch B.	200 OK	PASS
Expiry Alert Radar	Calculated days remaining into 90d, 60d, 30d, and expired buckets.	200 OK	PASS
Batch Verification	Searched BATCH001 -> VERIFIED; searched fake batch -> NOT FOUND.	200 OK	PASS
Cold-Chain Compliance	Insulin (2-8C): 5C logged COMPLIANT; 12C logged OUT_OF_RANGE.	201 CREATED	PASS
Insufficient Stock Block	Requested 1,000 units when only 25 units exist in verified stock.	409 CONFLICT	PASS
Duplicate Batch Block	Attempted to insert already existing batch number BATCH-PCM-001.	409 CONFLICT	PASS

Live Working Results from Implemented System

The application is live and running on localhost with realistic seeded hospital data:

Executive Dashboard

- Total Medicines: 5
- Active Batches: 10
- Low Stock Items: 4
- Expiring Soon (<90d): 4
- Expired Batches: 1

FEFO Dispense Console

Shows live preview queue of available verified batches. When staff dispenses, it generates a real-time transaction receipt showing remaining balances.

Cold-Chain Monitor

Thermal audit simulator tests live sensor temperatures against drug limits and displays red alert badges for out-of-range violations.

Verified API Test: Executed live curl and PowerShell REST tests verifying all 6 endpoints (/api/dispense, /api/batches/verify, /api/temperature-logs, /api/expiry-alerts, /api/dashboard/stats, /api/medicines).

Future Enhancements & Scalability Roadmap

These are planned future improvements to expand this academic project for real commercial hospital deployment:

1. Barcode / QR Scanning

Integrate camera-based 2D DataMatrix scanning to automatically read batch numbers and expiry dates from boxes.

2. Wireless IoT Sensors

Connect real IoT temperature sensors inside medicine refrigerators to broadcast telemetry automatically.

3. SMS / Email Alerts

Send immediate automated SMS alerts to the pharmacy manager when temperature excursions occur.

4. User Login & Roles

Implement Spring Security to provide separate logins for Pharmacists, Storekeepers, and Doctors.

5. Cloud Deployment

Deploy backend and database on AWS or Render to enable access across multiple hospital branch locations.

6. AI Demand Forecasting

Use machine learning models to forecast medicine demand trends based on seasonal illness patterns.

Conclusion & Viva Defense

Project Conclusion

- Successfully built a full-stack pharmacy supply chain simulator.
- Enforces strict FEFO dispensing to reduce drug expiry losses.
- Monitors cold-chain compliance to safeguard sensitive medicines.
- Protects against counterfeit batches with 1-click verification.

What We Learned

- Developing clean REST APIs using Java Spring Boot 3.
- Managing 3NF database persistence with Spring Data JPA.
- Creating responsive clinical user interfaces with HTML5 & CSS3.
- Wiring frontend & backend using JavaScript Fetch API.
- Systematic debugging of CORS and HTTP error status codes.

Thank You!

We are now ready to answer questions from respected examiners.